

BankSimFM:

A MarS-Inspired Retail Banking Financial Distress Early-Warning Simulator

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01

Background and Problem Statement



- I. Practical Dilemma
- II. Research Motivation
- III. Innovation Thesis and MarS Alignment

What is the **problem**?

- Banks identify distress **only after** negative events occur
- Banks **rarely combine signals** (salary timing, recurring obligations, spending pressure, credit utilization) to capture the path into distress
- Traditional scorecards and static rules **miss the temporal progression** before missed payments or overdrafts

Conclusion:

Shift from reactive response → proactive early warning



Research Motivation and Questions

- Core idea: Customer financial behavior is **naturally sequential**
- Salary → repayment ability → bills → liquidity → balance deterioration → rising risk
- Three core questions:
 - ① Lag: How to proactively identify customers likely to enter distress within 30 days?
 - ② Prediction: How to predict specific behavioral trajectories instead of simple risk scores?
 - ③ Decision support: How to evaluate different interventions before acting?

BankSimFM enables forward-looking early warning, trajectory prediction, and intervention assessment through causal sequence modeling

Innovation Thesis & MarS Alignment

Innovation Thesis:

- A financial foundation model can operate directly on customer-event sequences
- Learns richer short-term risk dynamics than static summary models

MarS → BankSimFM Mapping:

MarS concept	BankSimFM equivalent
Order sequence	Customer financial-event sequence
Market trajectory	Customer cash-flow and distress trajectory
Generative sequence model	Causal transformer over banking events
Clearing mechanism	Deterministic account-state engine
Controllable generation	Intervention-conditioned simulation
Market-risk detection	Financial distress early warning
What-if market analysis	Intervention scenario analysis

02

Data Strategy and Preparation



- I. Overall Data Strategy
- II. Prototype Data & Preprocessing Pipeline
- III. Event Schema & Distress Definition

Overall Data Strategy: Hybrid Approach

Four categories of production data:
Specify data needed for real bank deployment



Category	Fields	Purpose
Customer profile data	Income band, employment type, region, product holdings, tenure, risk segment	Customer segmentation and repayment capacity assessment
Account and credit state data	Balance, overdraft limits, credit limits, utilization, delinquency status	Liquidity and credit pressure assessment
Event stream data	Salary credits, transfers, rent, utilities, card spend, loan due, loan payments, missed payments, ATM withdrawals, fees, support contacts	Core input for model training
Outcome labels	Overdraft, missed payment, persistent low-balance flags, delinquency stages, 30-day distress label	Supervisory signal

Prototype Data & Preprocessing Pipeline

- **Preprocessing core steps:**

1. Sort by time → 2. Customer-level split (60/20/20) → 3. Bucketing → 4. Sequence truncation (256 events) → 5. Intervention augmentation (30% windows injected with future intervention tokens)

- **Training sample structure:**

Component	Example
Context window	Last 256 customer events
Event tokens	salary_credit -> rent_payment -> card_spend -> credit_card_payment_due -> card_spend
Dense features	balance, utilization, days to next due, due-amount feature, intervention flag
Primary targets	next event, next amount bucket, next balance-delta bucket, distress label
Intervention token	none, reminder, due_date_shift_7d, temporary_overdraft_buffer, or installment_restructure



Since real banking data is inaccessible, this project implements a synthetic data generator.



- **Five customer archetypes:**

- ① Stable salaried (low risk)
- ② Volatile income (medium-high risk)
- ③ High obligation (medium risk)
- ④ Rising utilization (high risk)
- ⑤ Near distress (very high risk)



- **Data scale:** 300 customers, 82,391 event rows

Event Schema & Distress Definition

- **Distress definition (any of the following within 30 days):**

- ✗ Missed loan or credit card payment
- ✗ Account overdraft
- ✗ Persistent low balance
- ✗ other deterioration signals

- **Data strategy assumptions:**

- ① Synthetic data reasonably approximates real customer behavior
- ② Customer-level split, no data leakage
- ③ 256 historical events sufficient to capture risk signals
- ④ Missing values handled via deterministic defaults

- **Each event contains both event and state information:**

Field	Description
customer_id	Synthetic customer identifier
event_timestamp	Time of event
event_type	Event class
amount	Event amount
amount_direction	Credit or debit
category	Merchant or event category
balance_before	Account balance before event
balance_after	Account balance after event
credit_limit	Credit limit
credit_utilization	Utilization ratio
days_to_next_due	Days to next scheduled repayment
intervention_flag	Intervention marker
distress_label_30d	Distress within next 30 days



03

Modern Architecture and training approach



3.1 Model Architecture: Main Idea



- MarS models order-level market trajectories; BankSimFM models customer-level banking event trajectories.
- Each customer history is treated as an ordered sequence of events, not a static profile.
- The model learns patterns before distress, such as low balance, high utilization, missed payments, and failed debits.
- The goal is both forecasting and risk scoring.

● Causal Event Transformer

- Inputs: event type, amount bucket, balance bucket, utilization bucket, due bucket, time-gap bucket, intervention type, and position.
- Transformer encoder captures relationships across the customer history.
- The final valid hidden state becomes the customer-level representation.
- Four output heads predict: next event, next amount bucket, next balance-delta bucket, and distress risk.



Training Approach

- Transformer uses a multi-task objective: event prediction, amount prediction, balance-delta prediction, and distress classification.
- LSTM is trained as a simpler distress-classification baseline.
- Both models use the same train / validation / test split for fair comparison.
- Validation AUC is used for checkpoint selection; final metrics are reported on the test set.



04



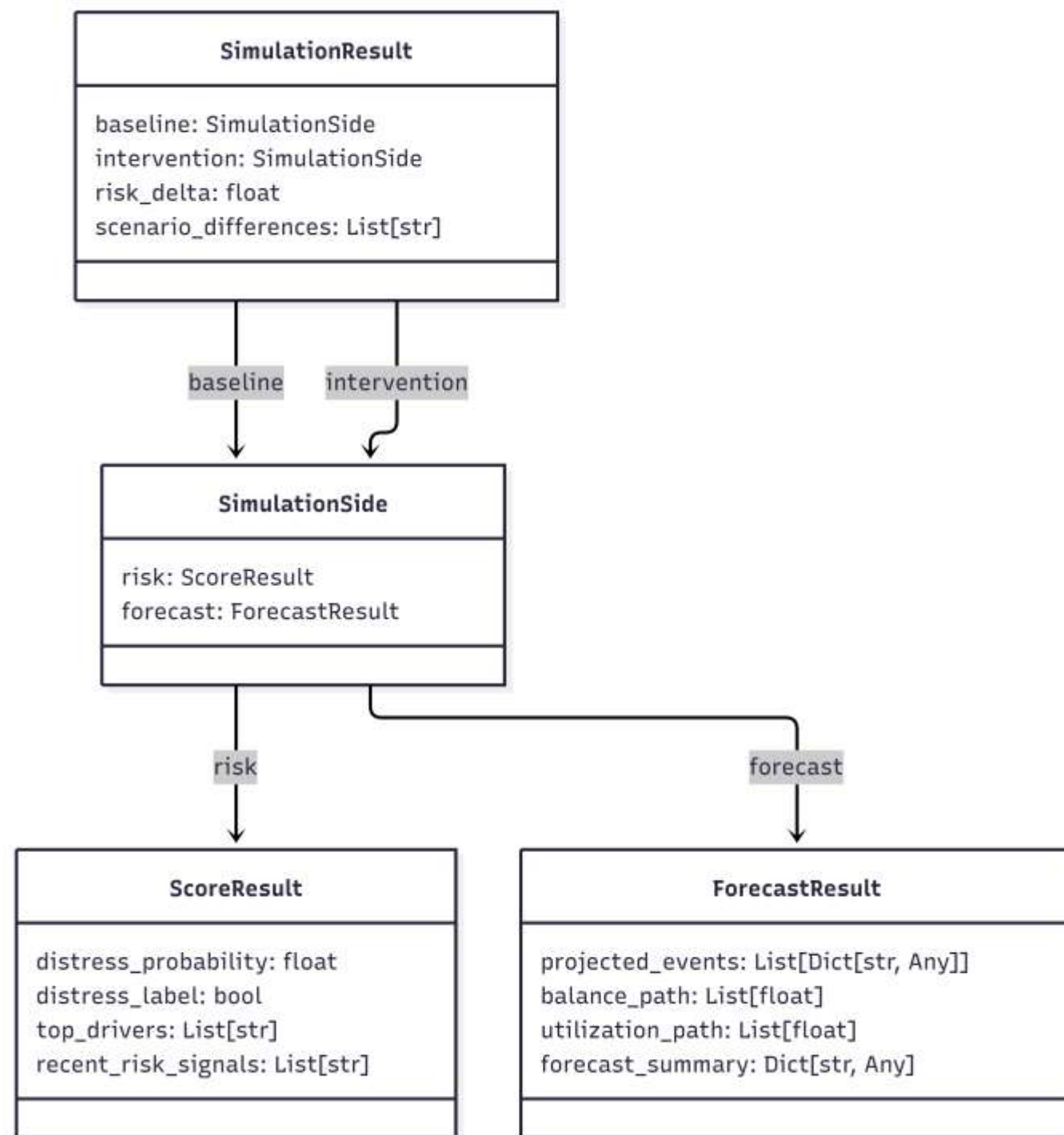
Forecasting, Socring, and Simulation Interfaces

4.1 Inference Interfaces

- `score_customer()`: estimates distress probability and returns key risk drivers.
- `forecast_customer()`: generates future customer events, balance path, and utilization path.
- `simulate_intervention()`: compares a baseline scenario with an intervention scenario.
- Together, these APIs turn the model into a dashboard-ready decision-support layer.

4.2 Forecasting and Scoring

- Customer history is cleaned, sorted by timestamp, and converted into model inputs.
- If trained artifacts are available, the Transformer is used as the main scorer.
- Forecasting uses the Transformer to project future events under the selected horizon.
- Outputs include distress probability, risk label, top drivers, recent signals, and forecast summary.



● Intervention Simulation

- The system first generates a baseline forecast using the original customer history.
- It then applies an intervention policy to the account state and runs a second forecast.
- The result compares baseline vs intervention risk, ending balance, utilization, and negative-event count.
- This supports intervention selection, collections prioritization, and portfolio monitoring.



05

Evaluation and Results



- I. Holdout Classification
- II. Early-Warning and Scenario Metrics
- III. Intervention Usefulness
- IV. Fairness and Segment Behavior

Holdout Classification & Early-Warning Results

Transformer vs LSTM:

Model	Test AUC	Test Precision	Test Recall	Test F1	Test Accuracy
Transformer	0.8342	0.6127	0.5556	0.5828	0.8696
LSTM	0.8126	0.6458	0.4250	0.5126	0.8675

- **Transformer** outperforms the LSTM in both AUC and F1 score
- LSTM achieves higher precision while slightly lower recall score
- Transformer outperforms LSTM in both AUC and F1 → **selected as primary live scorer**

Early-Warning Performance:

Metric	Value	Interpretation
Average lead time	21.55 days	weeks warning window
Hit rate on distressed customers	73.33%	Successfully identifies nearly 3/4 of distressed customers
False positive rate on stable customers	43.33%	Room for improvement in warning precision

Simulation Quality & Stability:

Metric	Value
Event-mix divergence	0.675
Utilization RMSE	0.0282
Ending-balance standard deviation	75.43
Negative-event-count standard deviation	0.257

Conclusion: The prototype generates meaningful scenario variation while maintaining financial consistency.

Intervention Usefulness

Intervention	Avg Predicted Risk Reduction	Share Improved By 5pp
due_date_shift_7d	0.0171	0.35
installment_restructure	<u>0.0228</u>	0.40
temporary_overdraft_b uffer	0.0217	0.35
reminder	<u>-0.0447</u>	0.35

- **Installment restructuring** is most effective (structural intervention)
- **Reminder** shows negative effect → not all interventions are equally effective
- Lightweight reminder offers limited help to stressed customers, and may even backfire
- **Conclusion:** Different customer segments require differentiated intervention strategies

Fairness and Segment Behavior

Evaluation dimensions: Customer archetype, income band, employment type, region, risk segment

Segment	AUC
High obligation	0.9175
Rising utilization	0.9105

- High-risk segments show strong performance
- Performance varies across regions and employment types
- Continuous subgroup monitoring required

➤ Final conclusion:

Transformer as primary scorer is justified. The prototype validates the feasibility of migrating MarS to retail banking.

06

Downstream Applications



- I. Early distress warning
- II. Intervention what-if analysis
- III. Portfolio stress monitoring
- IV. Collections prioritization

BankSimFM: Core Functions & Applications

❑ System positioning

- Not just a predictive model
- End-to-end decision system

❑ Four Core Functions

✓ Early Warning

- 3-week advance risk detection
- Proactive intervention window

✓ Intervention What-If Analysis

- "Simulate-first" approach
- Structural changes (e.g., debt restructuring) work best

✓ Portfolio Stress Monitoring

- Risk aggregation by customer segments
- Proactive risk identification

✓ Collections Prioritization

- Smart dashboard for analysts
- Focus on high-value cases



Translation into Operational & Economic Decisions

Application	Operational decision	Business value mechanism
Early warning	Review or contact customers earlier	Avoid missed-payment losses and downstream delinquency costs
What-if simulation	Select best support action	Improve intervention targeting and reduce wasted outreach
Portfolio monitoring	Watch vulnerable cohorts	Improve segment-level planning and staffing
Collections prioritization	Rank customers for action	Increase analyst productivity and intervention efficiency



07

Risk, Challenges & Governance



- I. Privacy
- II. Fairness
- III. Explainability
- IV. Reliability
- V. Controls



01



PRIVACY

Synthetic-only data lowers PII exposure; production still needs access, lineage, encryption, and retention controls.



02



FAIRNESS

Subgroup metrics surface uneven behavior, but synthetic segments do not guarantee real-world representativeness.



03

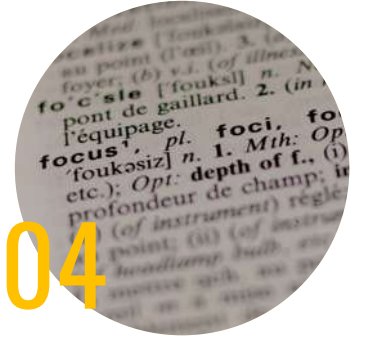


EXPLAINABILITY

Signals, driver heuristics, timeline review, and intervention comparisons give analysts useful context.



04



RELIABILITY

State-based simulation improves consistency, but drift, instability, and causal misuse remain risks.

08

Measurement of Economic Value



- I. Avoided losses
- II. Better targeting
- III. Analyst productivity
- IV. Retention benefit

08

VALUE
LOGIC

- I. **Identify At-Risk Customers:** Reduce missed payments and default losses early
- II. **Targeted Intervention:** Choose effective actions (e.g., restructuring) for higher ROI
- III. **Boost Analyst Productivity:** Smart ranking saves time and improves focus
- IV. **Retain Customer Value:** Prevent deterioration through timely support
- V. **Net Value:** Avoided Losses + Operational Savings
+ Retention Benefit – Action Cost

THANK YOU